

Meadowline Skin Co.: Profitability Diagnostic

Contribution economics of a DTC skincare store, July 2025 to June 2026 (USD)

Demo project based on synthetic Shopify data. Meadowline Skin Co. is a fictional brand. No real client, no real results. Every finding is about this dataset only.

1. Executive summary

\$283,485

Net revenue (P2-04)

\$160,981

Contribution (P3-002)

56.8%

Contribution / net
revenue (P3-003)

-\$12.45

Contribution per
order, discounted vs
no code, same order
type (CX1)

OBSERVATION The business is healthy at the contribution level. Margin is 56.8% of net revenue. The top 10% of buyers give 31.0% of contribution, so there is no dependence on a few big customers. In this dataset I did not find one large profit leak.

CALCULATION I found four places where economics differ a lot between segments. They are ranked below by how well the evidence holds up:

1. **Discounted orders earn less.** After matching on order type (new or repeat buyer, units, set or no set), a discounted order gives \$25.93 contribution and an order with no code gives \$38.38. The basket size at list price is the same (\$65.78 vs \$65.66). This does not prove discounts cause a loss. It shows the price of the discount per order.
2. **The welcome code (WELCOME15) costs \$10,821 and is linked to fewer repeat buyers.** 90-day repeat: 23.1% with the code, 29.9% without (CX4). The cause is unknown.
3. **Returns sit in a few products.** P06 Retinol has a 13.2% mature unit return rate. The all-products rate is 5.44%.
4. **TikTok does not repay its cost in 90 days.** 90-day contribution per new buyer is 0.69× CAC. Google shows 1.84×.

RECOMMENDATION All five recommendations are tests, not instructions. The data is observational, so it cannot prove cause.

2. Context

Meadowline Skin Co. is an invented direct-to-consumer skincare store on Shopify. The question for a founder is simple: where does money stay in the business, and where does it leak? The case study shows how Shopify data can answer this in a way that supports decisions.

3. Data and method

FACT Eight raw files: customers, orders, order lines, products, returns, shipping costs, discounts, marketing spend. Period: 2025-07-01 to 2026-06-30. Taxes are not in revenue. There are no operating expenses, so the supported metric is **contribution**, not net profit.

- Every number comes from code run on the files. Each number has an ID in a Number Register.
- Every statement is labelled: FACT, CALCULATION, OBSERVATION, HYPOTHESIS or RECOMMENDATION.
- Segments with fewer than 30 observations are called "insufficient evidence".

- Discount comparisons are matched on order type: new or repeat buyer, number of units, and whether the order has the set. Return comparisons are matched on product and first vs repeat order. I checked for Simpson's paradox. I found none.
- Recent orders had less time to be returned or repeated. I used mature groups only: orders up to 2026-05-16 for returns, first orders up to 2026-03-31 for 90-day repeat.
- Customers are counted once per email_hash.

4. Data quality

ISSUE	COUNT	DECISION
Orders in file	5,073	Valid orders: 5,006
Voided orders	61	Excluded (DQ-O02)
Test orders	6	Excluded (DQ-O03)
Duplicate order lines	11	Removed. They explained all subtotal mismatches.
Duplicate shipments	4	Removed
Orphan rows (returns / shipments / discounts)	4 / 3 / 2	Removed
Missing postage	—	Estimated from parcel weight. Total \$358.75 vs \$355.84 with a second method (DQ-S02, DQ-S03)
Persons (by email) / buyers	4,034 / 3,828	One person can have several customer IDs

OBSERVATION The errors look injected by the data generator. Real stores have different problems.

5. Revenue

STEP	USD	BASE
Gross sales	331,063.00	Quantity × list price, valid orders
Discounts	-32,423.89	9.79% of gross
Refunds	-15,154.37	5.07% of sales after discounts
Net revenue	283,484.74	85.6% of gross (P2-04)
Shipping charged to customers	19,069.05	Added in contribution

FACT Average order value: \$66.13 at list price, \$59.66 after discounts, \$56.63 net. 52.9% of orders carry a discount code.

OBSERVATION November 2025 is the peak: 634 orders and \$34,678.99 net revenue. Discounts were 21.5% of gross sales in November and 7.8% in the other 11 months. Jan-Jun 2026 had 25.7% more orders and 35.2% more net revenue than Jul-Dec 2025. The November peak may come from the generator.

6. Products

CALCULATION Contribution = net revenue + shipping charged – product cost (net of restocked returns) – postage – packaging – payment fees – return shipping. Order-level costs are split by gross line value. A split by units gave the same picture. Only 3 of 12 products change rank between revenue and contribution.

PRODUCT	NET REVENUE	CONTRIBUTION	MARGIN
P03 Vitamin C Brightening Serum 30ml (top contribution)	\$50,643	\$29,589	58.4%
P04 Hyaluronic Hydration Serum (highest margin)	\$32,536	\$20,176	62.0%
P11 Body Lotion 400ml (lowest margin)	\$10,491	\$4,588	43.7%

Margin = contribution / net revenue of the product. Company total: contribution costs are 27.4% product cost, 18.3% postage and packaging, 4.2% payment fees (all % of net revenue).

7. Discounts

CALCULATION CX1 compares discounted orders with orders without a code, inside groups of the same order type (new or repeat buyer × number of units × set or no set). 9 of 12 groups were usable (at least 30 on both sides), covering 98.3% of discounted orders.

	DISCOUNTED	NO CODE
Contribution per order (matched)	\$25.93	\$38.38
Same, simple average	\$26.51	\$38.77
Contribution margin	51.9%	61.0%
Basket at list price	\$65.78	\$65.66

Difference: –\$12.45 per order, 95% interval –13.34 to –11.53 (bootstrap).

HYPOTHESIS Customers who would have bought anyway take the discount. Then the discount is a pure margin cost. The data cannot test this. It needs a holdout test.

Repeat behaviour by first-order discount (CX2)

FIRST-ORDER DISCOUNT DEPTH	NONE	~15%	16–25%	≥26%
90-day repeat rate	30.5%	23.8%	17.7%	15.2%

OBSERVATION The pattern holds without Nov–Dec. It can come from who gets a deep discount, so it is not proof of effect.

Welcome code and BFCM30

- CALCULATION** WELCOME15: \$10,820.95 on 1,176 orders, 3.8% of net revenue. Break-even: 23.6% of users would need to be incremental (discount \$11.26, contribution \$36.34, 763 orders). This ignores the lower repeat rate.
- CALCULATION** 90-day repeat, WELCOME15 vs no code, same source: 23.1% vs 29.9% (–6.7 pp, interval –10.9 to –2.4). The code group is lower in 7 of 7 sources and without Nov–Dec. Unknown and unclear source labels

were excluded (294 of 2,740).

- **OBSERVATION** BFCM30-only orders in Nov–Dec: \$21.71 contribution per order (margin 42.8%) vs \$39.75 (60.4%) with no code. Mix was not controlled here.
- **FACT** Orders with both BFCM30 and WELCOME15: 30. Too few for a conclusion.

8. Returns

- **CALCULATION** Cost of returns: \$15,501.82, or 8.78% of contribution before returns.
- **FACT** 77.7% of refund dollars went to returnless refunds. Only 8.4% of returned units went back to stock.
- **FACT** Damaged or wrong-item reasons are 15.6% of return rows.
- **CALCULATION** Mature unit return rate: 5.44% overall. P06 Retinol 13.2% (interval 10.9–16.0%), P12 Set 9.0%, P02 1.5%. First orders return 6.1%, repeat orders 3.4%.
- **CALCULATION** P12, P06 and P03 give 65.0% of refund dollars from 45.0% of net sales before refunds (\$134,482 of \$298,639; P3-078).
- **CALCULATION** Discounted vs not discounted, matched by product and first/repeat (24 cells): 6.81% vs 4.22%, +2.59 pp (interval +1.52 to +3.60). For P06 first orders: 20.1% vs 8.5%. Excess refund dollars up to \$1,125.97 (association only, upper bound).
- **OBSERVATION** 42 of 91 P06 returns have reason "irritation_breakout".
- **CALCULATION** One point less return rate is worth \$264 for P06 and \$455 for P12, in refund dollars only.

HYPOTHESIS Return rates that differ so clearly by product may be generator settings. I would not call them a rule for skincare.

9. Customers

- **CALCULATION** Repeat buyers (two or more orders in the window): 22.7%. This is a lower bound, because earlier history is not in the data.
- **CALCULATION** Repeat orders are 23.5% of orders and 28.9% of contribution. Contribution per first order: \$29.89. Per repeat order: \$39.54.
- **CALCULATION** Top 10% of buyers give 31.0% of contribution.
- **CALCULATION** 90-day repeat rate: 24.2% on average, with a range of 19.7% to 30.0% between monthly groups. I do not claim lifetime value. I only report what was seen in the window.

10. Marketing

FACT Marketing spend: \$76,018.48. Meta is 50.3% of it. Spend is by channel and month. Source labels in the customer file are messy: 182 records were mapped (IG to Meta, "Google" to Google Ads). The channel order does not change in the strict or generous cases.

CHANNEL	CAC	90-DAY CONTRIBUTION / CAC	90-DAY CONTRIBUTION – SPEND
Google	\$21.73	1.84×	+\$6,255.07
Meta	\$36.05	1.12×	+\$3,149.58
Influencer	\$30.48	1.07×	+\$623.96
TikTok	\$47.89	0.69×	–\$3,037.09

CAC = channel spend / new buyers from that source. Ratio and difference use the Jul–Mar group, so each buyer has 90 days of data.

- **CALCULATION** 90-day repeat: TikTok 10.4%, Google 25.4%.

- **CALCULATION** Platform-reported ROAS is 2.67. In Shopify, first-order ROAS is 1.68, and 2.21 when all orders of those buyers count. The channel order is the same, but platforms report higher numbers.
- **CALCULATION** Total contribution minus all marketing: \$84,962.88 (total level only, before operating costs).
- **CALCULATION** Nov vs Oct 2025: orders +61.7%, spend +99.2%, contribution after marketing -14.8% (\$5,916.75 vs \$6,946.55).

HYPOTHESIS With 12 monthly points per channel I do not test correlations. Channel repeat rates may be generator settings.

11. Shipping (checked, not ranked high)

CALCULATION Shipping charged covers 37.0% of shipping cost. Net result: -\$32,518.86 (P3b-006, P3b-007). US orders over \$50 ship free: 77% of \$50-75 baskets, 99% of \$75-100 and 100% of \$100+ paid \$0. Orders bunch above \$50: 34 in \$45-50 vs 114 in \$50-55.

OBSERVATION This looks like a deliberate free-shipping rule that may raise basket size. I cannot call it a leak without a test. Upper limit of any saving, not a forecast: \$4,206.65 for \$50-75 baskets (707 orders at \$5.95) and \$8,568.00 for \$75+ (1,440 orders).

12. Key findings

F1. Discounted orders earn less (confidence: high on the gap, low on the cause).

Evidence: CX1. Impact: \$12.45 less contribution per discounted order at the same order type. Explanation (hypothesis): buyers who would purchase anyway. Next: holdout test.

F2. WELCOME15 costs \$10,821 and links to lower 90-day repeat (confidence: medium).

Evidence: CX2, CX4, break-even 23.6%. Explanation (hypothesis): the code attracts deal seekers. Next: test the code against a smaller or no code.

F3. Returns are concentrated in P06 and P12 (confidence: high on concentration, medium on cause).

Evidence: unit return rates, P3-078, CX3b. Dollar size is small (\$264 per point for P06). Next: check reviews, batches and usage guidance.

F4. TikTok does not repay spend in 90 days (confidence: medium).

Evidence: CAC, 90-day ratio, -\$3,037.09 gap. Limit: source labels are messy, and the window is short.

F5. Shipping recovery is 37.0% (confidence: high on the fact, low on whether it is a problem).

Evidence: P3b-006, P3b-007. Likely policy. Next: threshold test.

13. Recommendations

R1. Test WELCOME15 against a lower code.

Why: F1, F2. Evidence: CX1, CX4, break-even 23.6%. Expected effect: not estimated in dollars; the test measures the incremental share. Largest possible effect in this dataset: \$10,820.95 total code cost. How: random split of new visitors, 10-15% see 10% or no code, 60-90 days. Watch: contribution per new buyer after 90 days and repeat rate.

R2. Block stacking of BFCM30 with a welcome code.

Why: stacked orders reach 40.5% depth. Evidence: only 30 orders, so the loss is not shown. Expected effect: inconclusive. How: a rule in Shopify. Watch: share of orders with two codes and their margin.

R3. Investigate P06 before any price or discount change.

Why: 13.2% return rate, 42 of 91 returns for skin irritation. Evidence: CX3, CX3b. Expected effect: small in dollars, possible effect on trust. How: check usage text, batches, reviews. Do not remove the product on one metric. Watch: P06 mature return rate.

R4. Move 25% of TikTok spend to Google as a test.

Why: 0.69x vs 1.84x. Evidence: CAC and 90-day ratios. Expected effect: at most \$4,213.67 on \$3,663.43 moved, assuming Google keeps its average return. Real result will be lower. How: 6–8 weeks. Watch: CAC and 90-day contribution per new buyer by channel.

R5. Test the free-shipping threshold before changing it.

Why: F5. Evidence: P3b-006, P3b-007, order bunching above \$50. Expected effect: ceiling \$4,206.65 for \$50–75 baskets, not a forecast. How: A/B \$50 vs \$60. Watch: AOV, share of orders above the threshold, net shipping per order.

14. Assumptions and limits

ASSUMPTION	EFFECT
A1. Cost of set P12 = P01 + P03 + P07 = \$21.10	Below 1% of contribution
A2. Free gift P13 costs \$1.00 per unit. This is my placeholder, not store data. 277 units.	\$0–3 per unit changes contribution by ±\$554 (±0.34%)
A3. Missing postage estimated from weight	\$2.92 on total contribution
A4. Source labels IG and "Google" mapped to Meta and Google Ads (182 records)	Channel order unchanged
A5. Payment fee is not refunded on returns	Contribution is slightly understated
A6. Order-level costs split by gross line value	Product ranks change little (units-based check)

- The data is synthetic. Patterns come from a generator. Product return rates, channel repeat rates and the November peak may be artifacts.
- There are no operating costs. Contribution is not profit.
- The data is observational. No finding shows cause.
- Recent orders are excluded from return and repeat rates because they had less time.
- Platform spend and attribution are not checked against the platforms.

FACT QA: 24 of 24 checks passed. A separate calculation from the raw files gave contribution of \$160,984.28 vs \$160,981.36 in the main path. The \$2.92 gap comes from the postage estimate.

15. Metric definitions

- **Valid order:** not voided and not a test order.
- **Gross sales:** quantity × list price.
- **Net revenue:** gross sales – item discounts – refunds.
- **Contribution:** net revenue + shipping charged – product cost (restocked returns recovered, replacements added) – postage – packaging – payment fees – return shipping.

- **Contribution margin:** contribution / net revenue.
- **CAC:** channel spend / new buyers from that source in the same period.
- **Repeat (90-day):** share of new buyers with a second order within 90 days, first orders up to 2026-03-31.
- **Mature return rate:** returned units / sold units for orders up to 2026-05-16.

Demo project based on synthetic Shopify data. Tools: Python (pandas), code-run checks, Number Register.